

# Are Small LLMs Ready for Coding Agents?

A tight harness connects controllable models to accepted full workflows.

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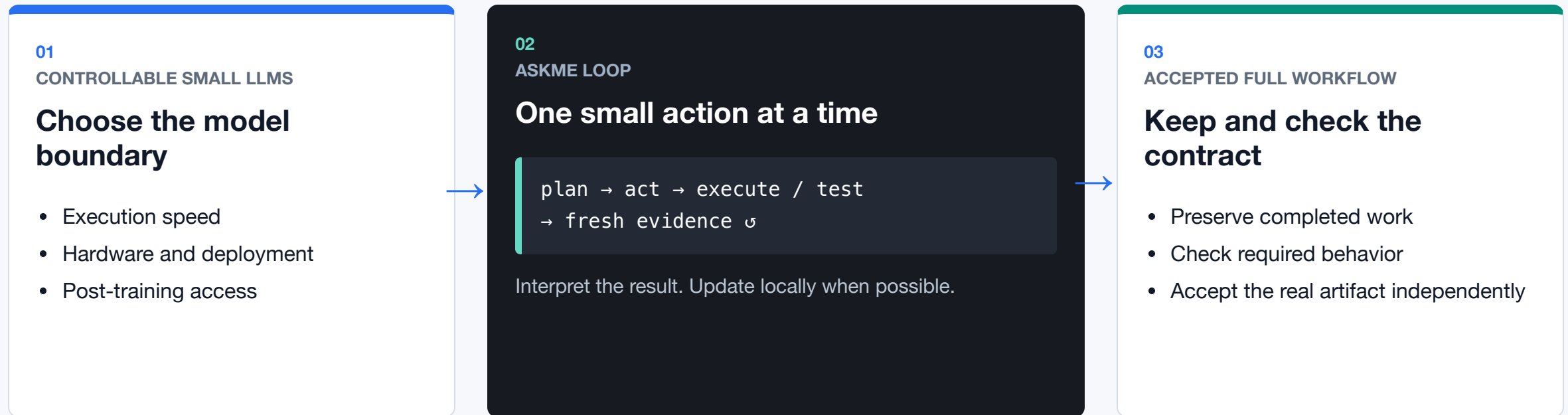
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# A tight loop connects model control to full workflows

The model gets one bounded move; the harness keeps the end-to-end contract.



**The bridge:** small action surface + fresh execution feedback + independent workflow acceptance.

# The code ran. The workflow still failed.

## REQUIRED CONTRACT

```
write msg.h + main.c  
build ./main  
run ./main  
observe REPLAN_OK
```

### Source and header written

The requested program was correct.

### `/tmp/test` compiled

A different output path was chosen.

### `/tmp/test` ran successfully

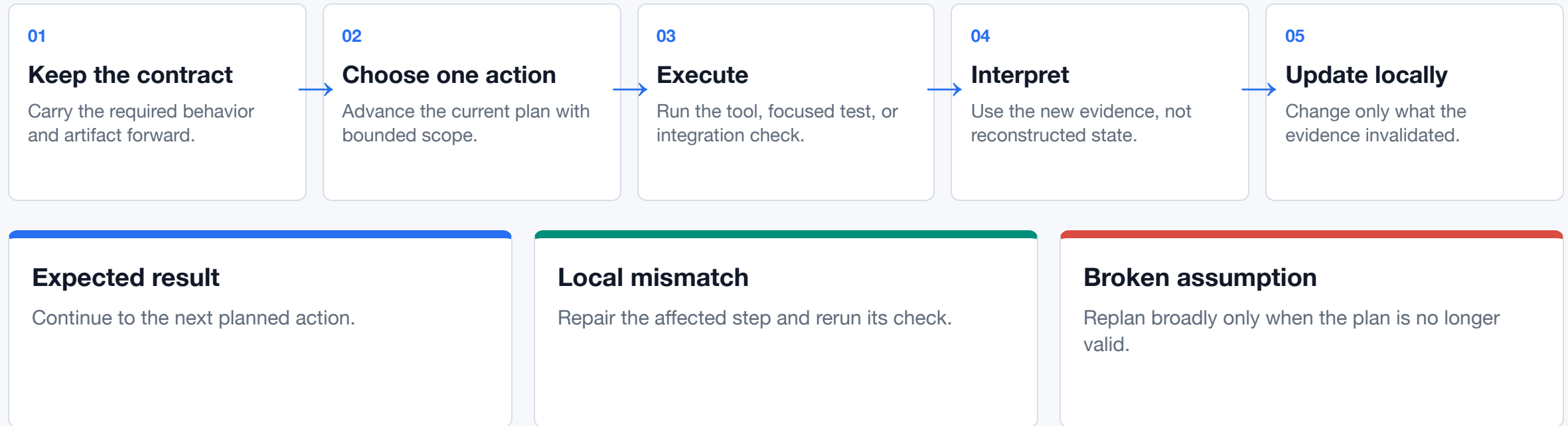
It printed `REPLAN_OK`.

### Agent reported completion

The independent acceptance test found no `./main`.

**Every recorded action succeeded.** The deliverable contract still drifted.

# Reason over fresh evidence. Preserve progress.



**Trajectory goal:** fewer repeated failures, fewer stuck steps, and less unnecessary plan churn—not longer monologues.

# Four models, eight descriptive receipts

Two variants per family; architecture and active parameters changed.

8 / 8 agent complete

7 / 8 artifact accepted

Hosted · simple checks · n=1/cell · descriptive only

| MODEL           | SHAPE                              | BUILD OBSERVATION                           | REPAIR OBSERVATION |
|-----------------|------------------------------------|---------------------------------------------|--------------------|
| Gemma 4 26B A4B | ♦ MoE<br>25.2B total · 3.8B active | ACCEPTED<br>603.6s                          | ACCEPTED<br>20.0s  |
| Gemma 4 31B     | • Dense<br>30.7B parameters        | ACCEPTED<br>66.5s                           | ACCEPTED<br>22.4s  |
| Qwen3.6-27B     | • Dense<br>27B parameters          | ACCEPTED<br>47.9s                           | ACCEPTED<br>23.0s  |
| Qwen3.6-35B-A3B | ♦ MoE<br>3B active                 | NOT ACCEPTED<br>wrong artifact path · 17.7s | ACCEPTED<br>11.8s  |

**Boundary:** architecture, active parameters, run order, and trajectories changed; the 31B follow-up was post-hoc. No size, family, reasoning, or reliability inference.

# Freeze the smallest clean test

01

## 4 frozen workflows

Valid multi-file code, semantic failures, visible feedback, held-out scoring.

02

## 2 frozen policies

Explicit-reasoning off versus the current composite gated policy.

03

## 3 randomized repeats

$4 \times 2 \times 3 = 24$  scheduled runs per model.

04

## 2 primary outcomes

Held-out acceptance and false completion across all valid runs.

**Secondary:** repeated actions · work redone · local/full replans · latency · tokens

ANSWER TO THE TITLE

# Promising—with a tight harness.

Model choice

Simple contract

Execution / test

Local reasoning

Accepted workflow

**Make the interface easier to use.  
Keep success tied to real behavior.**

The smoke validates this interface; readiness needs repeated held-out workflows.

[github.com/den-run-ai/askme](https://github.com/den-run-ai/askme) · slides, blog, protocol, and raw summary data